

A
DISSERTATION REPORT
ON
AUTOMATED CONTROL AND PROTECTION SYSTEM IN
INDUCTION MOTOR

Submitted to the
Sant Gadge Baba Amravati University
In the fulfilment of requirement of degree of
B.E. Electrical Engineering

Submitted By:
Mr. Arpit Sanjay Badhe (22BF310113)

Guided By:
Prof. Tejal Y. Kharche



Department of Electrical Engineering

LSSBM's

Padm. Dr. V. B. Kolte College of Engineering, Malkapur

Dist. Buldhana, 443101

2024-25

Vision and Mission of Institute

Vision

To be an autonomous institution offering multidisciplinary quality education programs aligned with provisions of NEP 2020 and satisfy stakeholder's needs.

Mission

1. To achieve excellence in academics, research and services continuously improving the quality.
2. To ensure the wholesome development of students implementing the provisions of NEP 2020.
3. To develop professional ethics, universal human values, constitutional values, and 21st century skills in students.
4. To observe provisions of quality education declared by the accreditation agencies.

Vision and Mission of Department

Vision

To provide high quality education and research in electrical engineering to satisfy the needs of industry and society.

Mission

1. To develop complex problem-solving competencies through laboratory work, projects and industrial training.
2. To create opportunities for the wholesome development of students to excel in a dynamic technical environment.
3. To inspire students to contribute for improving the quality of life of people in the vicinity.
4. To create facilities for research and development for electrical and allied engineering



LSSBM's

Padm. Dr. V. B. Kolte College of Engineering, Malkapur

Dist. Buldhana, 443101

Department of Electrical Engineering

CERTIFICATE

This is to certify that the dissertation entitled “**Automated control and Protection system in induction motor**” has been carried out successfully by **Mr. Arpit Sanjay Badhe**.

It is bonafied work carried out under supervision of **Prof. T. Y. Kharche** and is approved for partial fulfillment of requirement of university of Amravati, for the award of Degree of Bachelor of Electrical Engineering.

Prof. T. Y. Kharche

Assistance Professor
Electrical Engineering
VBKCOE, Malkapur

Prof. Y. P. Sushir

Head Of Department
Electrical Engineering
VBKCOE, Malkapur

Dr. A. W. Kharche

Principal
VBKCOE, Malkapur

Date:

Place: Padm. Dr. V. B. Kolte College of Engineering, Malkapur, 443101

EXAMINER'S CERTIFICATE

This is to certify that the dissertation report titled “**Automated Control and Protection System in induction motor**”, submitted by **Mr. Arpit S. Badhe** has been approved for the award of degree **Bachelor of Engineering** in Electrical Engineering of Sant Gadge Baba Amravati University.

Examiner

Name:.....

Guide

Prof. T. Y. Kharche

HOD

Prof. Y. P. Sushir

Principal

Dr. A. W. Kharche

Date:

Place: Padm. Dr. V. B. Kolte College of Engineering, Malkapur, 443101

ACKNOWLEDGMENT

I would like to express profound gratitude to my guide **Prof. T. Y. Kharche** for her cherished support, encouragement, supervision, and useful suggestions throughout this project work. Her moral support and continuous guidance enabled me to complete my project work successfully.

I am grateful for the cooperation and constant encouragement from our honorable Head of Department **Prof. Y. P. Sushir** His regular suggestions made my work easy and proficient.

I wish to express my appreciation to Principal of VBKCOE., **Dr. A. W. Kharche** who helps me to overcome my doubt in doing this project work.

Last but not the least, I am thankful and indebted to my family and all those who helped me directly or indirectly in the completion of this project work.

Name of student
Mr. Arpit Sanjay Badhe

Examination No.
22BF310113

ABSTRACT

Induction motors are used considerably domestically and industrially because of their ruggedness, low cost, ease of operation. On the other hand, they operate efficiently but demand high- position control and protection systems for dependable performance and safe operation. The principles and perpetration of control and protection systems for induction motors are bandied with respect to speed control, necklace control, fault discovery and forbearance, tackle-in- the- circle simulation (HILS), etc. Different control strategies like scalar and vector control ways are bandied, emphasizing the use of these styles in controlling the motor like a good performance at different cargo conditions. It's also studied the objectification of protection systems(overcurrent protection, thermal protection and phase- failure discovery) to avoid motor failures and increase its functional life.

It also explores the ultramodern technologies like microcontrollers grounded monitoring for the motor control and protection progress as well. With the help of these technologies, real-time monitoring and prophetic conservation can be fulfilled along with remote fault diagnostics making them a great improvement in effectiveness. In the end it concludes with some imperative directions and unborn trends making inductive motor intelligence systems an investment strategy in itself.

This begins by furnishing a foundational understanding of the induction motor, exploring its abecedarian principles, literal development, and crucial characteristics. latterly, it delves into a comprehensive disquisition of motor speed control ways grounded on optimization algorithms. Through a relative analysis, the strengths, limitations, and implicit operations of optimization-grounded control styles are illustrated, aiming to enhance the performance and effectiveness of induction motors across different artificial and marketable settings. First, this exploration aims to deliver a deep sapience into the crucial features of machine control and protection in induction motors for better artificial robotization results.

Keywords: Induction Motor, Automation, Protection, Control, Microcontroller, Fault Detection, Real Time Control.

INDEX

CH. NO	CONTENTS	PAGE NO
	Certificate	
	Examiner Certificate	i
	Acknowledgement	ii
	Abstract	iii
	List of Figure	vi
1	Introduction	1
1.1	Background	1
2	Literature Survey	3
2.1	Literature Review	3
2.2	Problem Identification	6
2.3	Motivation of Project	7
2.4	Objective of the project	8
2.5	Framework	8
2.6	Hypothesis	9
3	Overview of Project	10
3.1	Introduction	10
3.2	System Implementation data	10
	3.2.1 System architecture	10
	3.2.2 System Component	10
	3.2.3 Implementation process	11
3.3	Control mechanism	11
3.4	Protection Mechanism	12
3.5	Advantages	13
3.6	Disadvantages	14
3.7	Application	14

4		Details of Design	15
	4.1	Component Required	15
	4.1.1	Zero PCB	15
	4.1.2	Microcontroller	15
	4.1.3	7805 voltage regulator IC	16
	4.1.4	Single Phase Induction motor	17
	4.1.5	ACS712 Current sensor	18
	4.1.6	Temperature Sensor	18
	4.1.7	Tungsten Bulb	19
	4.1.8	Display with I2C	20
	4.1.9	SMPS Adapter	22
	4.1.10	Relay 12 Volt	22
	4.1.11	SW420 Vibration sensor	23
	4.1.12	HC05 Bluetooth Module	23
	4.1.13	Diode	24
	4.1.14.	Switch	25
	4.1.15	Resistor	25
	4.1.16	Power cable	26
	4.1.17	Inductive Choke	27
5		Block Diagram and Final Implementation	28
	5.1	Block Diagram	28
	5.2	Project Pictures	29
	5.3	Result	31
6		Conclusion and Future Scope	32
	6.1	Conclusion	32
	6.2	Future Scope	33
		Publication and Certificate	34
		Certificate	37
		References	40
		Program	44

LIST OF FIGURES

Sr. No	Figure No.	Figure Name	Page. No
1	1.1	Induction Motor	1
2	4.1	Zero PCB	15
3	4.2	Microcontroller Ic	15
4	4.3	7805 Voltage Regulator IC	16
5	4.4	Single phase induction motor	17
6	4.5	ACS712 Current Sensor	18
7	4.6	Thermistor	19
8	4.7	Tungsten Bulb	20
9	4.8	16x2 LCD Display	20
10	4.9	I2C	21
11	4.10	12-volt Adapter	22
12	4.11	12 Volt relay	22
13	4.12	SW420 Vibration sensor	23
14	4.13	HC05 Bluetooth Module	24
15	4.15	Diode	24
16	4.16	Switch	25
17	4.17	Resistor	26
18	4.18	Power Cable	26
19	4.19	Inductive Choke	27
20	5.1	Block Diagram	28
21	5.2	Project Pictures	29
22	5.3	Project Implementation	30

LIST OF TABLES

SR. NO.	TABLE NO.	NAME OF TABLE	PAGE NO.
1	5.1	Project Budget	31

CHAPTER 1

INTRODUCTION

Chapter 1 describes about the introduction of project named Automated control and protection system in induction motor.

1.1 Background:

The induction motor stands as a cornerstone of modern industrial processes, providing the driving force behind countless applications, from manufacturing to transportation. Its robustness, simplicity, and efficiency have solidified its position as the workhorse of the electromechanical world. Since its inception over a century ago, the induction motor has undergone numerous advancements, evolving to meet the demands of increasingly sophisticated systems and technologies. This introductory section aims to provide a foundational understanding of the induction motor, exploring its fundamental principles, historical development, and key characteristics. We delve into the operating principles that underpin its functionality, elucidating the interplay of magnetic fields and electromotive forces that drive its operation. Additionally, we trace the evolution of the induction motor, from its early beginnings to the present day, highlighting pivotal innovations and milestones that have shaped its trajectory.

Furthermore, this introduction sets the stage for a comprehensive exploration of motor speed control techniques based on optimization algorithms. By establishing a firm understanding of the induction motor, we lay the groundwork for analyzing the various strategies employed to regulate its speed effectively. Through this comparative analysis.



Fig 1.1 Induction Motor

we aim to elucidate the strengths, limitations, and potential applications of optimization-based control methods in enhancing the performance and efficiency of induction motors across diverse industrial and commercial settings.

In the realm of electromechanical systems, the ability to precisely regulate motor speed stands as a crucial factor influencing efficiency, performance, and overall system functionality. From industrial machinery and robotics to automotive propulsion and renewable energy systems, precise control over motor speed is essential for optimizing processes, conserving energy, and ensuring safe operation. This introduction serves as a gateway to the exploration of motor speed control techniques, with a focus on the application of optimization algorithms.

We embark on a journey through the principles, methodologies, and advancements in motor speed control, aiming to provide a comprehensive understanding of the underlying mechanisms and the diverse array of approaches employed to achieve desired speed profiles are taken into account.

Induction motors are the backbone of industrial operations, accounting for nearly 60–70% of all motors used worldwide due to their ruggedness, reliability, and simplicity of construction. However, their performance and longevity depend significantly on how well they are controlled and protected against various electrical and mechanical faults. As industries move towards automation and smart manufacturing (Industry 4.0), the demand for automated control and protection systems for induction motors has increased dramatically. These systems enhance operational efficiency, reduce downtime, and extend equipment life through real-time monitoring, adaptive control strategies, and fault diagnostics.

An automated control system manages the operation of the motor, including start/stop sequences, speed regulation, torque control, and synchronization with other machines. Traditional manual control methods have been largely replaced with sophisticated automation tools such as Programmable Logic Controllers (PLCs), Variable Frequency Drives (VFDs), and smart sensors. These systems enable precise motor control under varying load conditions and help optimize energy consumption, aligning with modern energy efficiency standards.

In parallel, protection systems are critical to safeguard the motor against faults such as overloads, phase imbalances, undervoltage, overvoltage, single-phasing, and overheating. Modern protection schemes utilize advanced electronic relays, real-time thermal modeling, and digital fault recorders. Innovations in fault detection algorithms, predictive maintenance using IoT (Internet of Things) technologies, and machine learning are reshaping how faults are anticipated and addressed (IEC 61850 Standard, 2020). The next chapter is literature review.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Review:

Mamdouh, M., et al. (2018): This paper addresses the critical need for ensuring the continuous and smooth operation of induction motors (IMs) in modern industries. It begins by examining traditional signal analyses, focusing on vibration and current signals as indicators of various IM faults. The study analyzes signals from faulty IMs in a laboratory setup and discusses the advantages and challenges of conventional approaches. Additionally, it provides an overview of research efforts in signal-based automation for condition monitoring methodologies, emphasizing statistical features, AI techniques, and training/testing strategies for fault diagnosis.

"Journal, I. (2020): This review explores the evolution of traction motor technology and its impact on mechanical health monitoring and fault diagnosis. It highlights motor current signature analysis as a technique for identifying harmonic signals in line current. Various faults affecting induction motors, such as rotor, stator, bearings, vibration, and air gap eccentricity, are examined alongside diagnosis techniques. The review emphasizes AI's role in advancing fault detection and provides a comprehensive overview of traction motor faults and diagnostic schemes, aiming to guide future exploration and innovation.

"Bonet-Jara, J., et al. (2021): This study evaluates the effectiveness of recently developed approaches for choosing weighting factors in finite control set pulse width modulation (PWM) techniques. It categorizes methods into offline and online categories based on how weighting factors are computed, focusing on the evaluation of online methods for their ability to adjust factors automatically. The study scrutinizes four newly devised methods and the conventional method, delineating their merits and limitations to aid practitioners in selecting optimal approaches.

"Jannati, M., et al. (2017): This paper discusses speed control of induction motors using variable frequency drives (VFDs). It provides a basic understanding of VFD concepts and operation, highlighting the need for variable speed operation in response to varying load requirements. The document explores methods for changing motor speed, emphasizing VFD-based techniques.

"Ouanjli, N. El, et al. (2019): This study investigates methods for estimating sensorless speed in induction motors, focusing on their applicability for diagnosis purposes. It offers a comprehensive analysis of commercial diagnostic devices and real-world examples to elucidate prevailing challenges and guide future research efforts.

"Hannan, M. A., et al. (2018): This paper addresses the efficiency and control of single-phase induction motors (SPIMs) in residential and commercial electricity consumption. It emphasizes the need for efficient control strategies to optimize SPIM operation, providing a thorough examination of variable speed control techniques tailored for SPIM drives.

"Subasri, R., et al. (2020) : This review focuses on enhancing the performance of conventional direct torque control (DTC) for induction machines. It evaluates modern techniques targeting torque and flux control, assessing their effectiveness in managing algorithm complexity, minimizing losses, and improving control performance.

"Talla, J., et al. (2018) : This study critically assesses control and optimization strategies for induction motor (IM) drives. It provides a comprehensive analysis of scalar and vector control techniques, including speed and V/f control, direct and indirect field-oriented control, and various optimization techniques, aiming to guide future research in advancing IM drive technology.

"Sobhi, S. et al. (2023): This paper addresses the need for condition monitoring systems tailored to small electric motors in industrial settings. It presents an ongoing project aimed at developing a machine-learning-based solution for fault detection in multiple small electric motors, utilizing both shallow and deep learning approaches.

"Gudiño-Ochoa, A. et al. (2023): This study investigates the interaction of AC drives with induction motors in interharmonic generation. It analyzes the impact of different AC drives on interharmonic generation and employs time- frequency spectral analysis to understand harmonic generation, offering insights into the interaction between drives and motor characteristics.

Vijayakumari K V et. al. 2015, This paper is based on the “Three phase motors protection” from various faults. This paper emphasis on various difficulties such as phase reversal in single and three phase systems, drop out of line, damage due to overload, failure of single phase there will be usually recommended to perform in complex environmental conditions. This includes in built cut-off frequency having several facilities together with timer in off condition and a starter which is automatic. This main circuit of this controller comprises of a circuit with power supply, a latch and a counter which is used for phase sequence, timer of a motor with on and off conditions.

Igbinoba, C.K. et. al. 2020, In this paper, This paper presents the design and construction of a system for the protection of three phase induction motor from external faults such as overvoltage, under voltage and single phasing using operational amplifiers. The operational amplifier used in this work is LM 324 and was configured in the comparator mode by setting one of the inputs at a reference voltage and while varying the other input according to the mains input supply. The mains input voltage from each of the phase was stepped down to 12 VAC, regulated and divided using a potentiometer and fed to the varying input of the operational amplifier.

Dr. M. Chakravarthy et. al. 2014, In this paper , protection for three-phase Induction Motor is enabled using Crouzet Millenium PLC. The PLC correlates the operational parameters of the 3-Ø Induction Motor with the desired values preset by the user in the PLC that continuously monitor the system and isolate it during fault conditions. Thus, PLC proves an effective tool in industrial control of electric drives.

Shital S. Kalbande et. al. 2017, This paper describes protection of three phases Induction Motor from single phasing phase reversal, over voltage, and under voltage. Due to this electrical fault the winding of motor get burned which produce insulation damage and thus mitigate the life time of Motor. This fault is inducing in Induction Motor due to variation in Induction Motor parameters. When three phase Induction Motor works regularly, it is compulsory to protect the motor from these unlikely faults. Three phase Induction Motor directly connected through the supply, if the supply voltage has sag and swell due to damage the performance of Motor is affected and in some cases winding is heated .

2.2 Problem identification:

Induction motor is the backbone for every industry. However like any other machine, they will eventually fail because of heavy duty cycle, poor grounding environment, installation and manufacturing factors etc. With escalating demand for reliability and efficiency, the field of fault diagnosis in induction motor is gaining importance.

If the fault are not devine, it may result in large revenue losses as well as pose threat to reliability and safety of operation. Three phase induction motor generally suffers from under voltage, over voltage, overheating, single phasing and phase reversal problems. When the three phase induction motor supply with higher voltage than is rated value then induction motor starts overheated.

Overcurrent Protection and Motor Damage:

Induction motors are prone to overcurrent situations, especially during startup or under fault conditions. The absence of an efficient protection mechanism may lead to overheating, insulation damage, and even complete motor failure. Overcurrent protection is crucial to prevent these conditions and to ensure the motor operates within safe electrical limits.

Motor Speed Control Challenges:

Precise control of motor speed is essential for applications where speed variation is needed, such as in pumps, fans, or conveyor systems. Traditional methods like varying supply voltage or frequency can be inefficient or costly. Inadequate or improper speed control can lead to suboptimal performance and reduced efficiency.

Voltage Fluctuations and Motor Protection:

Fluctuations or sags in supply voltage can cause motors to operate outside their specified voltage range, leading to inefficient performance, overheating, or even damage to the windings. Ensuring the motor is protected from such irregularities is important to avoid unplanned downtime and maintenance costs.

Phase Imbalance or Loss:

Induction motors are vulnerable to issues arising from phase imbalance or loss of one phase. If the protection system is not designed to detect such faults quickly, the motor could experience excessive heating, reduced performance, or even permanent damage to the winding insulation.

2.3 Motivation of the project:

2.3.1 Project Motivation:

Automated Control and Protection System in Induction Motor. Induction motors find application in industries and within commercial arena for the simple reasons that they are robust, highly efficient, and require low maintenance. Unfortunately such motors are likely to produce internal faults such as overvoltage, undervoltage, overheating, overcurrent and phase imbalance, to mention a few, leading to reduced efficiencies, unplanned downtimes, and costly repairs.

2.3.2 Main Motivation:

➤ Enhancement of Motor Protection:

- Conventional protection systems are based on either mechanical devices or hand-operated control and would thus lack the necessary speed to avert damage.
- Installation of an automatic control and protection system would provide the chance for real-time observation.

➤ Increased Efficiency and Reliability:

- In such a way, the scheduled monitoring of motor parameters as continuous voltage and current as well as the temperature and vibration source maintains the conditions for operating optimally.
- The early detection of faults helps to save energy and improves the lifespan of the motor.

➤ Decreased Downtime and Maintenance Expenditure:

- Spare motors can halt production and entrap very heavy losses due to unplanned failure.
- Automated systems can identify and diagnose faults before they develop and recommend corrective action that can be taken before the failure occurs, thereby minimizing repair costs.

➤ Safety Enhancement:

- Electrical faults in induction motors may create dangerous scenarios through fire hazards and damage to equipment.
- An automated protection system optimally minimizes these types of risks thereby ensuring a safe environment to work in.

2.4 Objective of the project:

The Main objectives of dissertation are as follows:

- Design of control system
- Design of Protection system
- Fault Detection and Protection
- Safety and reliability Enhancement
- Optimization of motor Performance
- Testing and validation
- Implementation of communication protocols
- To use voltage sensors to identify the voltage required.
- To use current sensors to identify the current required.

2.5 Framework of the project:

Chapter 1 includes the introduction of project and induction motor. Chapter 2 literature survey which includes some sub topic like literature review, problem identification, motivation of project, objective of project, framework, hypothesis.

Chapter 3 overview of project which includes sub topic introduction, implementation process, control mechanism and protection mechanism, Advantages and Application.

Chapter 4 includes Details of design which includes components used in project prototype. Components used like microcontroller, ACS712 Current sensor, Temperature sensor, Sw420 Vibration sensor, HC05 Bluetooth module, I2C, Display.

Chapter 5 Block diagram and final implementation which includes subtopic like block diagram, description, project pictures, Project Budget.

Chapter 6 conclusion and future scope which includes conclusion of selected project and future scope of project. After chapter 6 contains are reference, publications and program used.

2.6 Hypothesis:

H₀: Null Hypothesis:

"An induction motor in operation with or without such an automated control and protection system will not show any significant improvements in its efficiency, reliability, and fault prevention as compared to one having such a system."

H₁: Alternative Hypothesis:

"Introducing an automated control and protection system to an induction motor will lead to greatly improved efficiency, reliability, and life span by avoiding faults such as overcurrent, thermal overloads, voltage variations, or phase imbalance in real time."

Sub-head Hypotheses:

Fault Detection Efficiency:

H₀: The system does not improve early fault detection in induction motors.

H₁: The system improves early fault detection by continuously monitoring key motor parameters.

Motor Performance and Reliability:

H₀: Motor performance and reliability are independent of the use of the automated system.

H₁: Motor performance is improved through the use of the automated system, preventing failure and thereby reducing unplanned downtime.

CHAPTER 3

OVERVIEW OF PROJECT

Chapter 3 describes about the overview of project entitled automated control and protection system in induction motor.

3.1 Introduction:

Induction motors are widely used in industrial, commercial, and residential applications due to their robustness, efficiency, and low maintenance. However, to ensure their reliable operation, appropriate control and protection systems are essential. The control system ensures the motor runs optimally under various load conditions, while the protection system prevents damage due to faults such as overload, short circuits, and voltage fluctuations. This paper provides a comprehensive overview of the control and protection systems in induction motors, including their working principles, components, and applications.

However, to ensure their optimal performance and to prevent potential damage from electrical faults or mechanical failures, a comprehensive control and protection system is essential. The objective of this project is to design, implement, and analyze a robust control and protection system specifically for induction motors.

3.2 System implementation data:

3.2.1 System Architecture:

The automated control and protection system consists of three major components:

- Sensors: Collects real-time parameters like voltage, current, temperature, and speed.
- Microcontroller: Processes sensor data, applies control algorithms, and triggers protective actions.
- Actuators and Display Interface: Controls motor operation and provides feedback via display or alerts.

3.2.2 Components of the system:

➤ Sensors:

- Current Sensors: Measure motor current to detect overcurrent conditions.
- Voltage Sensors: Monitor input voltage to prevent under/overvoltage.

- Temperature Sensors: Detect overheating conditions.
- Vibration Sensors: Identify mechanical faults and imbalance.
- Microcontroller:
 - Processes sensor data using embedded software.
 - Executes protection algorithms and sends control signals.
 - Interfaces with Human-Machine Interface (HMI) for real-time monitoring.
- Protection and Control Mechanism:
 - Overcurrent Protection: Trips motor if current exceeds threshold.
 - Overvoltage/Undervoltage Protection: Shuts down in case of voltage anomalies.
 - Overtemperature Protection: Initiates cooling or motor shutdown.
 - Speed Control Mechanism: Uses Pulse Width Modulation (PWM) for efficient speed regulation.

3.2.3 Implementation process:

- Sensor Installation: Place sensors at critical motor points.
- Microcontroller Programming: Develop firmware to process sensor data and execute control logic.
- Communication Setup: Integrate IoT modules for real-time monitoring.
- Testing and Calibration: Verify system response under normal and fault conditions.
- Deployment and Monitoring: Install the system in an industrial environment and analyse performance.

3.3 Control mechanism:

3.3.1 Start/Stop Mechanism:

Start-stop control is a fundamental requirement for any induction motor system. It ensures that the motor operates only when required, thus improving efficiency and reducing wear and tear.

Manual Start-Stop Control:

This method involves push buttons and contactors. The basic circuit consists of:

Start Button (NO - Normally Open): When pressed, it energizes the contactor coil, closing the main contacts and starting the motor.

Stop Button (NC - Normally Closed): When pressed, it de-energizes the contactor coil, stopping the motor.

3.3.2 Reverse/Forward Control:

Reverse-forward control allows an induction motor to rotate in both directions, which is crucial in conveyor systems, lifts, and machine tools.

Manual Reverse-Forward Control

Uses two separate contactors, one for forward rotation and the other for reverse rotation.

Interlocking Mechanism: Ensures that both contactors are not energized simultaneously, preventing short circuits.

3.3.3 Speed Control:

Speed control of induction motors is essential in various applications, from conveyor belts to industrial machinery. Speed can be controlled by:

Voltage Control

Achieved using autotransformers or thyristor-based voltage regulators. Suitable for small induction motors but inefficient for large motors due to high power loss.

3.4 Protection Mechanism:

3.4.1 Overload Protection:

Overload occurs when an induction motor operates beyond its rated capacity for an extended period. This leads to excessive current draw, resulting in overheating of motor windings, reduced efficiency, and possible failure. The causes of overload include excessive mechanical load, voltage fluctuations, blocked ventilation, and prolonged starting conditions.

3.4.2 Overvoltage Protection:

Over-voltage protection is critical in an automated control and protection system for induction motors. Implementing advanced protective measures such as voltage monitoring relays, surge protectors, and automatic voltage regulators ensures motor safety, longevity, and efficiency.

3.4.3 Over Temperature Protection:

Overtemperature protection is a vital aspect of automated induction motor control and protection systems. By integrating advanced temperature monitoring technologies and automation strategies, industries can enhance motor efficiency, prevent unexpected failures, and reduce maintenance.

3.4.4 Vibration Protection:

Integrating vibration protection into an automated control and protection system for induction motors is crucial for ensuring efficiency, safety, and longevity. By leveraging real-time monitoring, predictive maintenance, and automated response mechanisms, industries can significantly mitigate the adverse effects of vibration, leading to improved productivity and reduced operational costs.

3.4.5 Automatic Power Factor Correction:

Automatic Power Factor Correction (APFC) is essential for maintaining energy efficiency and reducing power costs. By using APFC panels, industries and commercial buildings can optimize their electrical power usage, reduce penalties, and improve the overall reliability of their power system.

3.5 Advantages:

- **Energy Efficiency:** Automated control systems, particularly through the use of VFDs, can significantly reduce energy consumption by adjusting the motor's speed according to the load demand, thus preventing unnecessary energy waste.
- **Increased Lifespan:** By reducing the wear and tear on the motor (through smoother starts, stops, and speed transitions), automated systems can extend the operational life of induction motors.
- **Precise Operation:** Automated systems can ensure the motor operates at optimal speed, torque, and efficiency, leading to improved overall performance.
- **Enhanced Reliability and Uptime:** APS continuously monitors and safeguards the motor, reducing the likelihood of unexpected failures and minimizing downtime.

3.6 Disadvantages:

➤ **High Initial Cost:**

Implementing automated control and protection systems requires expensive sensors, controllers, and protection relays, increasing the initial setup cost.

➤ **Complexity in Design and Implementation:**

The integration of automation technology requires specialized knowledge and expertise. Advanced programming and calibration are needed for proper functionality. System troubleshooting and maintenance become more complicated compared to traditional methods.

➤ **Dependence on Sensors and Electronic Components:**

The system heavily relies on sensors (temperature, current, voltage) and electronic components. Failure of a single sensor can lead to false alarms or motor shutdown, impacting productivity.

➤ **Vulnerability to Electrical Noise and Interference:**

Industrial environments often have electromagnetic interference (EMI), which can disrupt signals in automated control circuits.

➤ **Maintenance and Downtime Issues:**

Unlike traditional relay-based protection, automated systems require periodic software updates and recalibration.

3.7 Application:

- **Mining and Metals Processing:** Motors that drive crushers, mills, and other heavy machinery.
- **Water Treatment Plants:** Motors driving pumps and compressors in water treatment systems.
- **HVAC Systems:** Induction motors control fans, pumps, and compressors, with automation ensuring optimal airflow and energy used.

CHAPTER 4

DETAILS OF DESIGN

4.1 Components Description:

4.1.1 Zero PCB:

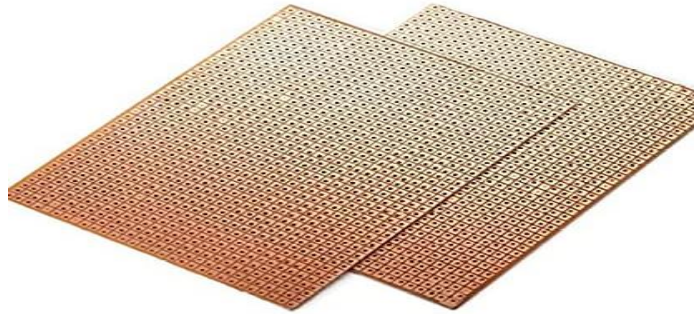


Fig 4.1 Zero PCB

Zero PCB means Printed Circuit Board, which is an all-purpose pre-drilled circuit board that is used to prototype electronic circuits. It has not been etched but has an array of holes with copper pads into which components can be soldered and interlinked through jumper wires or solder bridges. General purpose PCB as its name suggest, general purpose PCB are widely used to embed randomly for running hardware.

Features:

- Drilled holes in a uniform grid pattern where one could easily place components.
- Copper pads surrounding the holes for easy soldering.
- Ideal for prototyping before making a custom PCB.
- Best used for miniature projects as well as testing circuits without wanting to fabricate an entire PCB.

4.1.2 Micro-Controller:

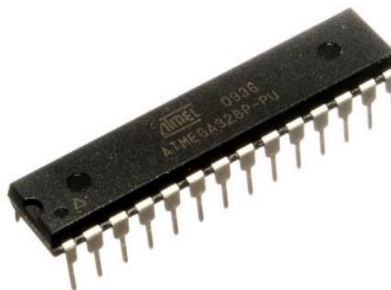


Fig 4.2 Atmega 328P Micro-Controller

The ATmega328P is an 8-bit microcontroller developed by Microchip Technology (formerly Atmel). It is a widely used microcontroller in embedded systems, particularly in Arduino boards, due to its power efficiency, versatility, and ease of programming.

The microcontroller is based on the AVR RISC (Reduced Instruction Set Computing) architecture, making it highly efficient in processing instructions per clock cycle.

Memory Specifications:

- Flash Memory: 32 KB (for storing programs)
- SRAM: 2 KB (temporary data storage)
- EEPROM: 1 KB (for non-volatile data storage)
- I/O and Peripheral Features:
- Digital I/O Pins: 23 (for general-purpose input/output operations)
- PWM Channels: 6 (used for analog output control, motor control, and LED dimming)
- Analog Inputs: 6-channel 10-bit ADC (Analog-to-Digital Converter)
- Timers: 3 (one 16-bit and two 8-bit timers for precise timing operations)
- Interrupts: External and internal interrupt support

4.1.3 7805 Voltage Regulator IC

The 7805 voltage regulator is a linear voltage regulator IC used by many to convert higher input voltage into fixed output 5V DC. It is part of the 78xx series in which "xx" becomes the output voltage. When it comes to power supply circuits, the 7805 is most often going to be found in circuits designed for ensuring stable voltage supply to microcontrollers, sensors, and other electronic components. It is the top choice primarily because it is simple and reliable while also having build-in safety features.

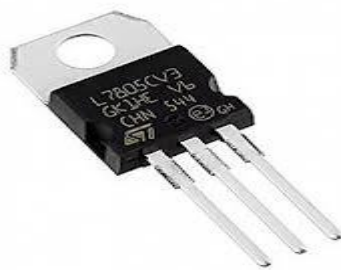


Fig 4.3 7805 Voltage Regulator IC

General Specifications:

- Output Voltage: 5V (fixed)
- Input Voltage Range: 7V to 35V
- Maximum Current Output: 1A (can be increased using a heatsink)
- Dropout Voltage: ~2V (minimum input must be at least 7V)
- Line Regulation: 3mV to 50mV
- Load Regulation: 15mV to 100mV

4.1.4 Single phase Induction Motor:

An induction motor or asynchronous motor is an AC electric motor in which the electric current in the rotor that produces torque is obtained by electromagnetic induction from the magnetic field of the stator winding. An induction motor therefore needs no electrical connections to the rotor. An induction motor's rotor can be either wound type or squirrel-cage type.



Fig 4.4 Single Phase Induction Motor

Construction:

The stator of an induction motor consists of poles carrying supply current to induce a magnetic field that penetrates the rotor. To optimize the distribution of the magnetic field, windings are distributed in slots around the stator, with the magnetic field having the same number of north and south poles. Induction motors are most commonly run on single-phase or three-phase power, but two-phase motors exist; in theory, induction motors can have any number of phases.

Advantages:

- Simple construction and low cost
- Reliable operation with minimal maintenance

4.1.5 ACS712 Current Sensor:

The ACS712 is a Hall effect-based current sensor widely used for accurate and reliable current measurement in both AC and DC applications. There are low-cost sensors customarily made by Allegro Microsystems, which are small in size, and they find applications in industrial, commercial, and consumer domains for current monitoring and protection.

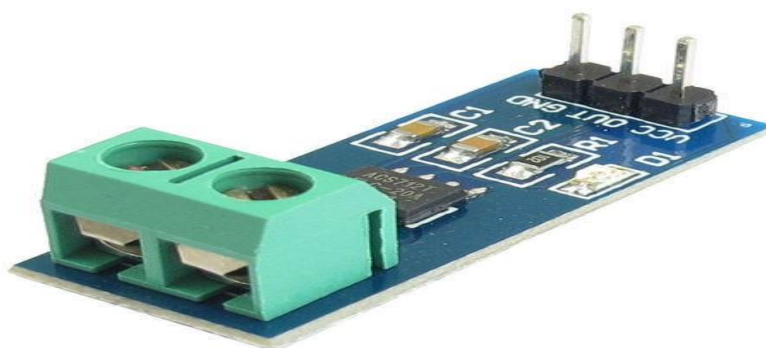


Fig 4.5 ACS712 Current Sensor

Operation:

The ACS712 works on the principle of the Hall effect, which states that when current is passed through a conductor, and the conductor is placed in a magnetic field, a voltage (Hall Voltage) gets generated perpendicular to the flow of current and the field. The ACS712 sensor consists of an integrated Hall-effect sensor and a copper conduction path along which current is flowing. The sensor can sense the magnetic field produced by the current and output a voltage which is proportional to that magnetic field.

4.1.6 Temperature Sensor (Thermistor):

The thermistor is a semiconductor type resistor with very high temperature dependence of resistance. The very word thermistor is the fusion of the two words thermal and resistor. Because of their variable resistance with temperature, these devices can be employed either as temperature sensors or to control current based on temperature. Some thermistors have a resistance that decreases with temperature, while other types have the opposite effect, i.e. an increase in resistance with increased temperature.

This gives rise to limiting current to cold circuits, e.g. for inrush-current protection, or limiting current to hot circuits, e.g. to prevent thermal runaway.



Fig 4.6 Thermistor

Construction:

Thermistors are normally made from metal oxides. They are typically pressed into beads, disks, or cylindrical shapes and encased in an impermeable material such as epoxy or glass.

NTC thermistors are produced from the oxides of iron-group metals; e.g. CrO, Cr₂O₃(chromium), MnO, CoO, iron oxides(nickel), NiO, and Ni₂O₃. These oxides constitute a ceramic body with terminals made of conductive metals mainly, silver, nickel, and tin.

PTCs are prepared from barium, strontium, or lead titanates, like PbTiO₃.

Applications:

- In Temperature Sensing and Measurement: Used in medical thermometers, weather stations, and household appliances.

4.1.7 Tungsten Bulb:

An incandescent light bulb/lamp/globe is an electric light with filament heated until it glows, the filament is mounted in a glass bulb which is either evacuated or filled with an inert gas to prevent oxidation of the filament.

These bulbs come in many sizes, outputs of light, and voltage ratings from about 1.5 to 300 volts. They need no external equipment for regulation, have low manufacturing costs, and work just as well on both AC and DC current.



Fig 4.7 Tungsten Bulb

Construction:

An incandescent light bulb consists of an air-tight glass enclosure (the envelope, or bulb) with a filament of tungsten wire inside the bulb, which receives the passage of electric current. Contact wires and a base with two (or more) conductors provide electrical connections to the filament.

4.1.8 Display (16x2 with I2C):

Display (16x2):

A liquid-crystal display (LCD) is basically a flat panel display or an optical device modulated electronically that uses the light-controlling ability of liquid crystals and polarizers for information display. Liquid crystals do not actively generate light but use a backlight or reflector to display colored and monochromatic images.



Fig 4.8 16 x 2 Display

Key Features:

- Format: 16 chars and 2 lines.
- Controller: Usually, based on the Hitachi HD44780 or its compatible versions.
- Interface: 4-bit or 8-bit parallel communication.
- Operating Voltage: 5V, and some modules also support 3.3V.

- Backlight: Available in different colors (e.g., blue, green, white).
- Character set: Standard ASCII characters and custom symbols.
- Power Consumption: Very low, which makes them ideal for battery-operated devices.

Inter-Integrated Circuit(I2C):

An Inter-Integrated Circuit (I2C) presents it as a serial protocol that has been developed specifically to enable communication between multiple integrated circuits of a system. I2C was initially developed by Philips Semiconductor in the 1980s for data communication between microcontrollers, sensors, memory chips, and any other devices for less complex wiring. I2C is mainly employed in very small embedded systems, consumer electronics, and most industrial applications because it requires only two wires to communicate to many devices. This is quite different from the SPI and the UART, which require many chip-select lines since it uses an addressing mechanism.



Fig 4.9 Inter-Integrated Circuit

I2C features:

- It is the multi-master and multi-slave: Unlike SPI, which only deals with a single master device at a time, I2C can have several masters on the same bus.
- Addressing scheme; Addresses for every single chip connected to the bus can be dedicated with a unique address of 7 bits or 10 bits. Thus, identification of the devices is very efficient.
- Bidirectional Communication: This includes full duplex communication, that enables both master and slave devices to transfer and receive.
- Clock Synchronization and Arbitration: It ensures the resolution of conflicts in cases where multiple masters usually want to control the bus at the same time.

Applications of I2C:

I2C is mainly used in very many areas, such as Embedded Systems: In here is used to connect a micro-controller to a sensor, EEPROM, and real-time clocks.

4.1.9 SMPS Adapter:



Fig 4.10 SMPS 12 volt Adapter

This adapter can be termed as an introduction to the 12-volt adapter. The 12-volt adapter is a simple power converting machine, which converts electrical energy from an alternate current (AC) power source to precise direct current (DC) output voltage of 12 volts. It is commonly used in almost all electronic industries, whether automotive, household, or industrial. The adapters became very popular because of their use in ensuring stable continuous energy supply to appliances requiring 12V DC.

4.1.10 Relay 12 Volt

A relay is an electrical switch. It has input terminals for control signals and operating contact terminals. A switch may include an arbitrary number of contacts in one of the contact configurations of make contacts, break contacts, or combinations thereof. A relay device is something where one circuit is controlled by an independent low-power signal or where multiple circuits are controlled by one signal.



Fig 4.11 Relay 12 volt

Operation:

A simple electromagnetic relay consists of a coil of wire wrapped around a soft iron core (i.e. a solenoid), an iron yoke that provides a low reluctance path for the magnetic flux, a movable iron armature, and one or more sets of contacts (there are two contacts in the relay depicted here). The armature is hinged to the yoke and mechanically linked to one or more sets of moving contacts.

4.1.11 SW420 Vibration Relay:

The SW-420 Vibration Sensor Module is a cheap and adaptable vibration sensor module that can be employed in a variety of situations. It is widely used in security applications, industrial monitoring, automotive applications, and amateur projects.



Fig 4.12 SW420 Vibration Sensor

Specifications:

- Operating Voltage: 3.3V-5V
- 15mA of current is consumed by it.
- Output type: Digital (High/Low)
- Sensitivity adjustment by means of a 10K Potentiometer
- Connectivity: VCC, GND, D0

4.1.12 HC-05 Bluetooth Module:

The HC-05 Bluetooth module has proven itself to be affordable and one of the most popularly used Bluetooth communicators enabling wireless serial communication between a microcontroller with a computer or any other Bluetooth device.

It supports the Bluetooth 2.0 with EDR (Enhanced Data Rate), making it a great option for projects involving Arduino, Raspberry Pi, and other embedded systems.



Fig 4.13 HC-05 Bluetooth module

The HC-05 Bluetooth module is known for its compatibility with Bluetooth version 2.0+EDR (Enhanced Data Rate) for wireless data transmission between devices. For ease of use and affordability, it is commonly used in hobbyist electronics projects, relying on specifications such as default baud rate of 9600, simplistic pinout, and customizing.

Features:

- Bluetooth Standard: Bluetooth V2.0+EDR
- Operating Voltage: 3.6V to 6V (Normally: 5V)
- Operating Current: 30mA to 40mA
- Communication: Serial UART (Rx,Tx)
- Baud Rate: Default 9600 (configurable)
- Operating Range: Up to 10 m (33 ft)

4.1.13 Diode:

A diode is an electronic component with two terminals through which electric current can flow in one direction. Diodes are the most basic constituents of electronic circuits and have many uses, including rectification, signal enhancement, voltage regulation, and various types of protection circuits. A diode is made from semiconductor materials, principally silicon or germanium, which create a junction of p-n that determines the property of current flow through the diode.



Fig 4.14 Diode

Structure and Working Principle:

- A diode is formed with two semiconductors: the p-type (positively doped) and the n-type (negatively doped). These two are joined to form the junction p-n. The characteristics of the diode depend on the behavior of this junction:
- Forward Bias: In this case, a higher potential is connected to the p-side while the n-side is connected to a lower potential. Thus, the diode allows the flow of current.
- Reverse Bias: The applied voltage is reversed; thus, it prevents the passage of current with very little leakage current.

4.1.14 Switch:

An AC switch is an electrical device that controls the flow of AC current in a circuit. Such switches can be used in the household application, industrial application, and the power distribution system.



Fig 4.15 Switch

Operation of Switch:

AC switch operating principle works by making or breaking an electric connection in a circuit. When ON switch is turned on, closing the internal contacts to allow the current flow. When the switch is turned OFF, the current contacts open, shutting off the current flow.

In these types of solid-state switches, semiconductor components like thyristors, triacs, or MOSFETs are used to operate these switches. Thus, for switching, It is much faster than the mechanical ones and more reliable since they have no mechanical components.

4.1.15 Resistors:

Resistors are passive components in electrical and electronic circuits because they do not produce energy but dissipate it in the form of heat. These devices are made primarily to limit current flow and divide voltages across circuit components. Resistors find their application in power management, signal conditioning, and circuit protection.



Fig 4.16 Resistors

Parameters:

These are some of the main parameters that define a resistor's performance:

- Resistance (Ohms, Ω): This is the measure against current limiting.
- Tolerance (%): This is the acceptable variation from the nominal value of the resistance.
- Power Rating (Watts, W): the maximum power it can dissipate without damaging the part.
- Temperature Coefficient (ppm/ $^{\circ}\text{C}$): this relates change in resistance to changing temperature.
- Voltage Rating (V): Maximum voltage which a resistor can withstand.

4.1.16 Power Cable:

Power cables play a very important role in electrical and electronic systems as they transmit and distribute electrical energy. These cables can be applied to various purposes including household wiring, industrial power distribution, and telecommunication networks.



Fig 4.17 Power cable

Construction of Power Cables:

- Conductor: The conductor is responsible for carrying the electric current and it is preferably made of copper or aluminum.
- Insulation: The electrical isolation of the conductors from the surrounding environment is provided by insulation. The most common insulations are PVC, XLPE, and rubber.

Applications of Power Cables:

Power cables find applications in almost every sector, for example:

- Residential and Commercial Buildings: for internal wiring and Lights and Power Distribution.
- Industrial Plants: connecting machines to the electricity grid for power supply; Control system inputs and outputs are linked to power cables.

4.1.17 Inductive Choke:

An inductive choke: Very much defined within the background of an inductive component is a passive electrical device that tends to impede high-frequency alternating current and allows lower frequencies and direct current to flow through it. This is a type of inductor that's one of the most important devices being used in electrical circuits in filter applications, noise suppression, and power management.



Fig 4.18 Inductive Choke

Working Principles:

Chokes inductively work on Faraday's Law of Electromagnetic Induction, which states that a variable instantaneous current through a coil produces a magnetic field that opposes the change of current. The property hence so possessed is termed as inductive reactance, which is in fact increasing along with reiteration of frequency in addition to that it is so effective for high-frequency signals while allowing low-frequency signals or DC.

Construction and Design:

An inductive choke typically consists of:

- Core Material: Chokes use a core made of iron, ferrite, or air to enhance inductance.
- Winding: Copper wire is wound around the core to form a coil, with the number of turns
- Shielding: Some chokes include shielding to minimize electromagnetic interference with nearby components.

CHAPTER 5

BLOCK DIAGRAM AND FINAL IMPLEMENTATION

5.1 Block Diagram:

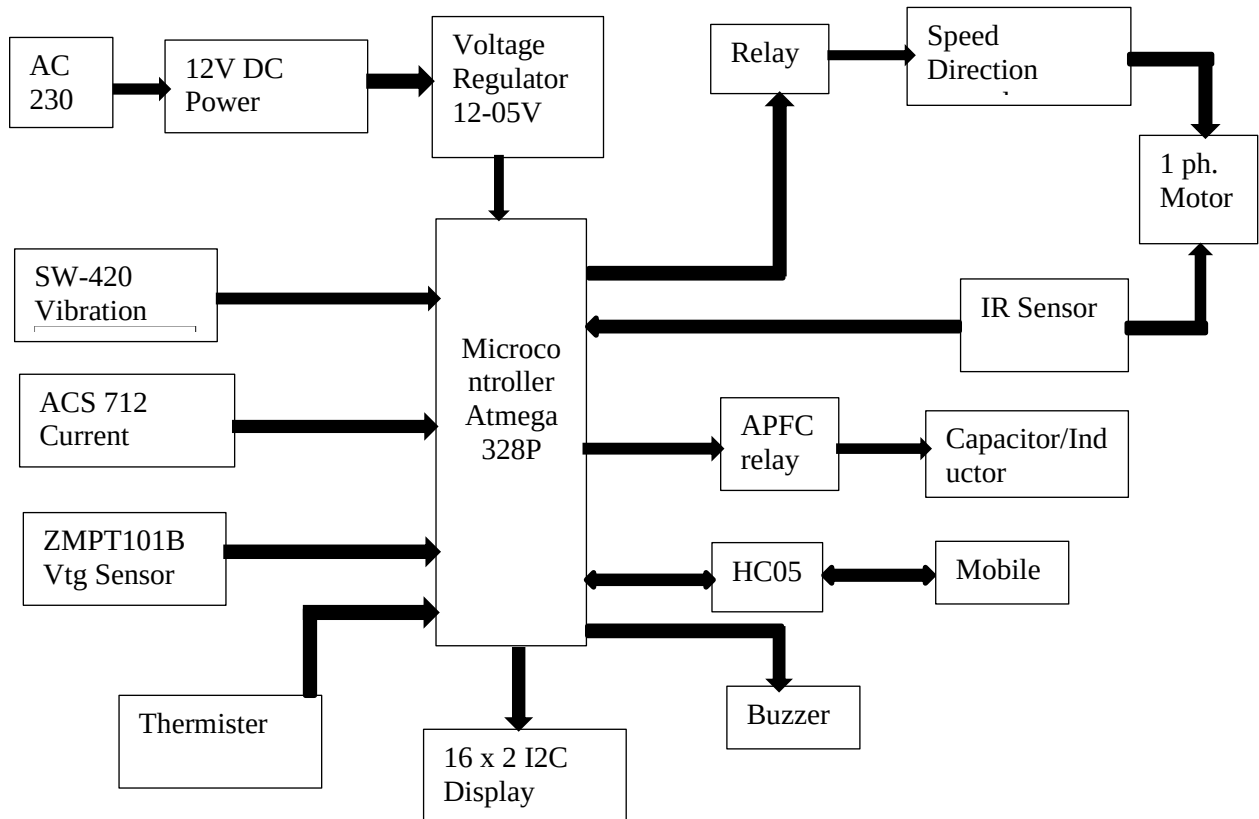


Fig 5.1 Block Diagram

Description Of Block diagram:

230 Volt Ac supply is used to start Model. 12 Volt Dc Converted from AC 230V using Adapter. Used Voltage regulator IC 7805 for convert to 05 Volt for operate Microcontroller IC ATMEGA328P and from this step we make power section for project.

SW420 Vibration Sessor for searching vibration fault, ACS712 Current Sensor for overcurrent fault, ZMPT101B Voltage Sensor, Thermistor for Overtemp. Fault these sensors are connected to Microcontroller ATMEGA328P for sensing data and transmit data and from this we make sensing section for our project.

16 x 2 with I2C Display. 16x2 LCD Display is used to monitor parameters. And I2C is used to operate LCD Display properly. 16 pins of LCD is connected to I2c and 4 pins of I2C is connected to ATMEGA328P IC and From this we make Display section for Project.

HC05 Bluetooth Module is used to transmit sending data to mobile and receive data from mobile and sending to microcontroller ATMEGA328P for Operation and from this we make Transmitting Section for project.

An IR sensor is used to Receive data of Rotation of induction motor. It receive data and transmit to Microcontroller and from this we observe rotation of motor.

5.2 Project Pictures:

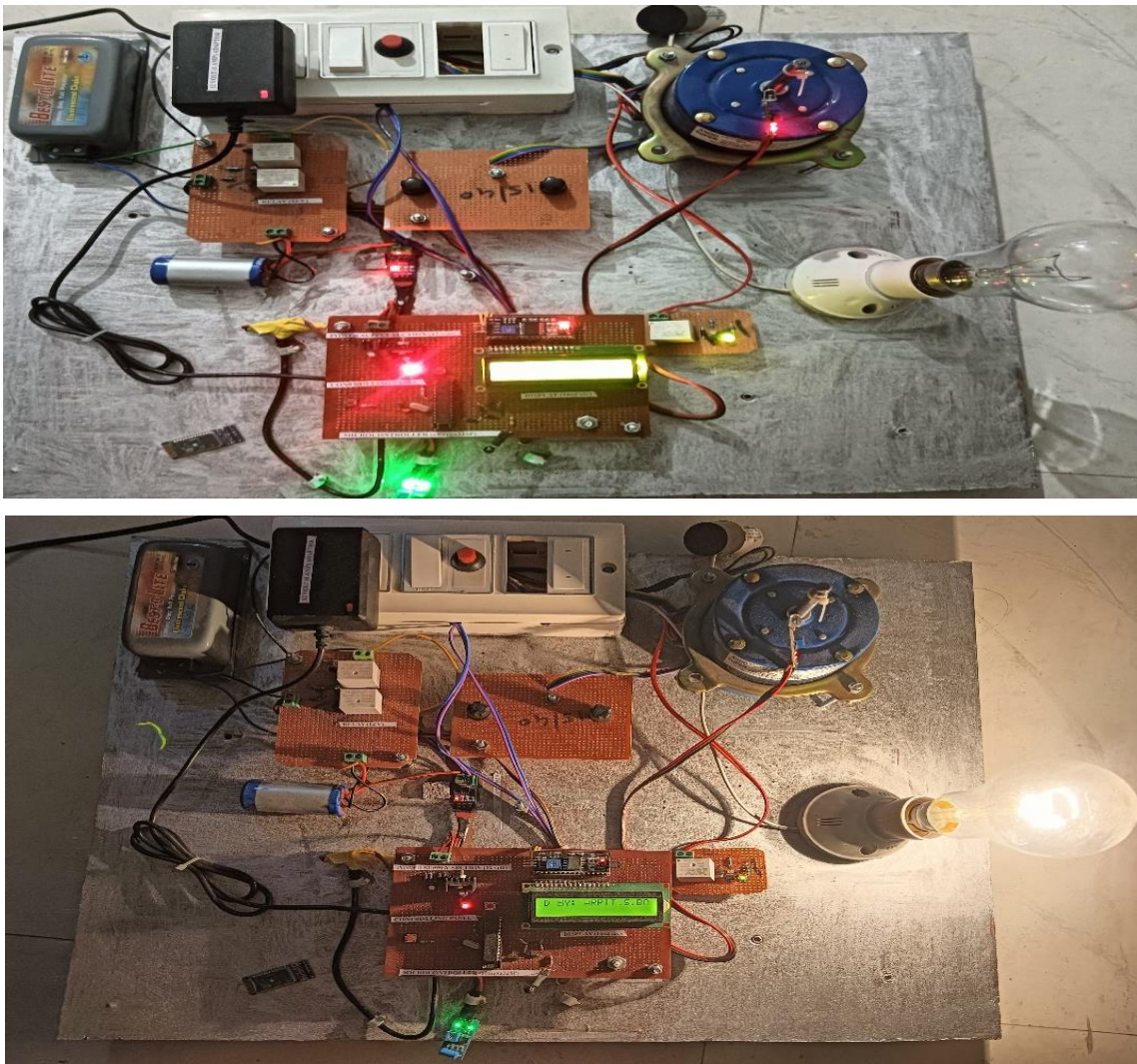


Fig.5.2. Final Implementation

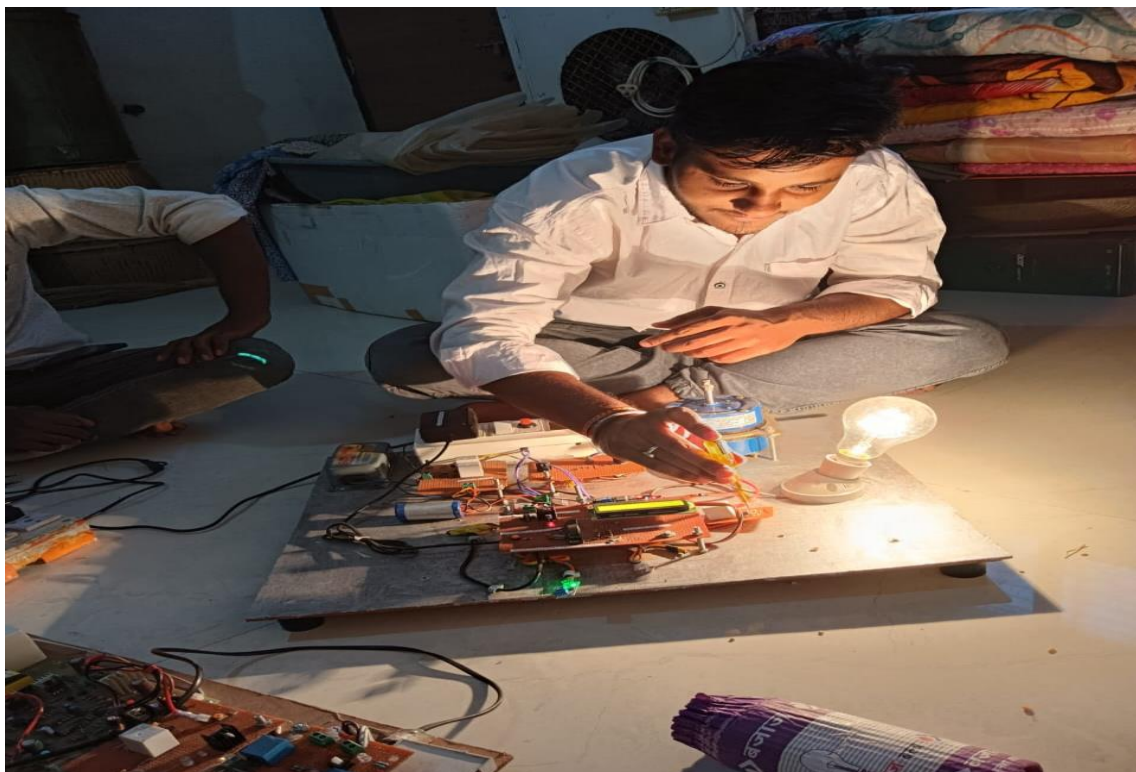


Fig 5.3 Project Implementation

5.3 Result:

The Temperature of motor is measured using Thermistor and displayed o LCD. Also send to the android app.If temperature is increased above certain limit, motor is stopped immediately to prevent from further damage. And also buzzer is turned on.The vibration of motor is measured using vibration sensor.If vibration is increased above certain limit, motor is stopped immediately to prevent from further damage. The motor is runs on overcurrent. Its measured through current sensor an its shows on display and send to mobile through HC05 module. Also Auto power factor correction is used.

5.4 Project Budget:

Table 5.1 Project Budget

Category	Description	Approximate cost	Remark
Material and Supplies	Black Book, Reports, Papers	Rs. 7,000	Printing, Stationary, Paper Publishing
Equipment's	Tools, Sensors	Rs. 10,000	Various sensor used, Motor, and other
Travel and Transport	-----	Rs. 1000	For Equipment Import
External Services	Third Party Services	Rs. 2000	Expert advice
Miscellaneous	Unexpected or minor expenses	Rs. 3000	Unplanned costs, and other
Total Approximate Expenditure		Rs. 23,000	

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion:

The road toward enhancing quite an important technical step for the automated control and protection schemes is focused on increasing induction motor efficiency, reliability, and safety for industrial and other commercial applications. Induction motors are widely used across sectors for their ruggedness, cost-effectiveness, and low maintenance requirements. However, the operation of induction motors is not without various fault risks: overvoltage, overcurrent, thermal overload, phase mismatching, and insulation fiasco are just a few of these. Hence, it's all the more important to have an automated control and protection scheme to avert possible damage to the motor and prolong its life expectancy.

Automated control schemes may monitor and control motor performance in real-time as the prime benefit. Conventional methods of control and monitoring may no longer be sufficient due to human errors and delayed response. Modern automated systems are further assisted by microcontrollers, programmable logic controllers (PLCs), and IoT that can sense and analyze motor parameters of interest such as current, voltage, temperature, and speed. All this data and information can enhance predictive maintenance, thereby minimizing downtime and saving costs.

Equally important, an automated protection scheme does not hesitate to act in fault situations instantaneously. Advanced protection schemes such as overcurrent relay, thermal sensors, and differential protection detect abnormal conditions and work without human interference. Such proactive measures would avert disasters and improve the general efficiency of the system. Further, fault diagnosis aided through artificial intelligence (AI) and ML algorithms has brought greater reliability to fault detection, enabling timely notifications and preventive maintenance actions.

Furthermore, the automation introduced into motor control applications provides efficient energy utilization. Induction motors consume a substantial portion of electrical energy worldwide.

6.2 Future Scope:

- The future of induction motor control and protection will focus on increased automation and intelligence.
- Integration with smart grid technologies with smart grid technologies will enhance energy management capabilities.
- Continuous innovation will drive improvements in efficiency, reliability and sustainability
- Wireless technologies are evolving, allowing for more flexible and scalable control and protection systems. Induction motors can be controlled and protected remotely, without the need for physical wiring or complex infrastructure.
- As industries evolve and new standards emerge, future control and protection systems will need to be developed in line with updated regulations regarding safety, energy efficiency, and environmental impact.
- As the demand for compact and affordable control systems increases, future developments could focus on miniaturizing protection and control components while maintaining high reliability and performance. This would be particularly beneficial in small-scale or residential applications.

PUBLICATIONS AND CERTIFICATES

1. Prof. T. Y. Kharche, Arpit. S. Badhe, Vikas. S. Sapkal, Aniket. V. Phirke, Ganesh. V. kakde, Vijaykumar. D. Dokhale, “Automated Control and protection system in induction motor,” December 2024 | IJIRT | Volume 11 Issue 7 | ISSN: 2349-6002
2. Prof. T. Y. Kharche, Mr. Arpit Badhe, Mr. Vikas Sapkal, Mr. Aniket Phirke, Mr. Ganesh kakde, Mr. Vijaykumar Dokhale, “Automated Control and Protection system in Induction Motor” March 2025 | IJPREAMS | Volume 05 Issue 03 | ISSN: 2583-1062



Certificate of Publication

The Board of International Journal of Innovative Research in Technology (ISSN 2349-6002) is hereby awarding this certificate to

PROF. T. Y. KHARCHE

In recognition of the publication of the paper entitled

AUTOMATED PROTECTION AND CONTROL SYSTEM IN INDUCTION MOTOR

Published in IJIRT (www.ijirt.org) ISSN UGC Approved (Journal No: 47859) & 7.37 Impact Factor

Published in Volume 11 Issue 7, December 2024

Registration ID 170407 Research paper weblink <https://ijirt.org/Article?manuscript=170407>


EDITOR


EDITOR IN CHIEF



Certificate of Publication

The Board of International Journal of Innovative Research in Technology (ISSN 2349-6002) is hereby awarding this certificate to

ARPIT SANJAY BADHE

In recognition of the publication of the paper entitled

AUTOMATED PROTECTION AND CONTROL SYSTEM IN INDUCTION MOTOR

Published in IJIRT (www.ijirt.org) ISSN UGC Approved (Journal No: 47859) & 7.37 Impact Factor

Published in Volume 11 Issue 7, December 2024

Registration ID 170407 Research paper weblink <https://ijirt.org/Article?manuscript=170407>


EDITOR


EDITOR IN CHIEF



Certificate of Publication

The Board of International Journal of Innovative Research in Technology (ISSN 2349-6002) is hereby awarding this certificate to

VIKAS SUDHAKAR SAPKAL

In recognition of the publication of the paper entitled

AUTOMATED PROTECTION AND CONTROL SYSTEM IN INDUCTION MOTOR

Published in IJIRT (www.ijirt.org) ISSN UGC Approved (Journal No: 47859) & 7.37 Impact Factor

Published in Volume 11 Issue 7, December 2024

Registration ID 170407 Research paper weblink <https://ijirt.org/Article?manuscript=170407>


EDITOR


EDITOR IN CHIEF



Certificate of Publication

The Board of International Journal of Innovative Research in Technology (ISSN 2349-6002) is hereby awarding this certificate to

ANIKET VIJAY PHIRKE

In recognition of the publication of the paper entitled

AUTOMATED PROTECTION AND CONTROL SYSTEM IN INDUCTION MOTOR

Published in IJIRT (www.ijirt.org) ISSN UGC Approved (Journal No: 47859) & 7.37 Impact Factor

Published in Volume 11 Issue 7, December 2024

Registration ID 170407 Research paper weblink <https://ijirt.org/Article?manuscript=170407>


EDITOR


EDITOR IN CHIEF





**International Journal of
Innovative Research in Technology**
An International Open Access Journal Peer-reviewed, Refereed Journal
www.ijirt.org | editor@ijirt.org An International Scholarly Indexed Journal

Certificate of Publication

The Board of International Journal of Innovative Research in Technology
(ISSN 2349-6002) is hereby awarding this certificate to

GANESH VASUDEO KAKDE

In recognition of the publication of the paper entitled

**AUTOMATED PROTECTION AND CONTROL SYSTEM IN INDUCTION
MOTOR**

Published in IJIRT (www.ijirt.org) ISSN UGC Approved (Journal No: 47859) & 7.37 Impact Factor

Published in Volume 11 Issue 7, December 2024

Registration ID: 170407 Research paper weblink: <https://ijirt.org/Article?manuscript=170407>

EDITOR

EDITOR IN CHIEF



**International Journal of
Innovative Research in Technology**
An International Open Access Journal Peer-reviewed, Refereed Journal
www.ijirt.org | editor@ijirt.org An International Scholarly Indexed Journal

Certificate of Publication

The Board of International Journal of Innovative Research in Technology
(ISSN 2349-6002) is hereby awarding this certificate to

VIJAY KUMAR DEVRAO DOKHALE

In recognition of the publication of the paper entitled

**AUTOMATED PROTECTION AND CONTROL SYSTEM IN INDUCTION
MOTOR**

Published in IJIRT (www.ijirt.org) ISSN UGC Approved (Journal No: 47859) & 7.37 Impact Factor

Published in Volume 11 Issue 7, December 2024

Registration ID: 170407 Research paper weblink: <https://ijirt.org/Article?manuscript=170407>

EDITOR

EDITOR IN CHIEF



**International Journal of Progressive Research in
Engineering Management and Science**
(Int Peer Reviewed Journal)

Ref: I/Certificate/Volume 05/Issue 03 /50300028336

Impact Factor: 7.001

DOI: <https://www.doi.org/10.58257/IJPREMS39047>

e-ISSN: 2583-1062

Date: 18/03/2025

Certificate of Publication

This is to certify that author "Prof. T. Y. Kharche" with paper ID
"IJPREMS50300028336" has published a paper entitled "AUTOMATED
CONTROL AND PROTECTION IN INDUCTION MOTOR" in International
Journal of Progressive Research in Engineering Management and Science
(IJPREMS), Volume 05, Issue 03, March 2025

Managing Editor

IJPREMS Journal

www.ijprems.com



Indexing services:



....



**International Journal of Progressive Research in
Engineering Management and Science**
(Int Peer Reviewed Journal)

Ref: I/Certificate/Volume 05/Issue 03 /50300028336

Impact Factor: 7.001

DOI: <https://www.doi.org/10.58257/IJPREMS39047>

e-ISSN: 2583-1062

Date: 18/03/2025

Certificate of Publication

This is to certify that author "Mr. Arpit Badhe" with paper ID
"IJPREMS50300028336" has published a paper entitled "AUTOMATED
CONTROL AND PROTECTION IN INDUCTION MOTOR" in International
Journal of Progressive Research in Engineering Management and Science
(IJPREMS), Volume 05, Issue 03, March 2025

Managing Editor

IJPREMS Journal

www.ijprems.com



Indexing services:



....



**International Journal of Progressive Research in
Engineering Management and Science**
(Int Peer Reviewed Journal)

Ref: I/Certificate/Volume 05/Issue 03 /50300028336

Impact Factor: 7.001

DOI: <https://www.doi.org/10.58257/IJPREMS39047>

e-ISSN: 2583-1062

Date: 18/03/2025

Certificate of Publication

This is to certify that author "Mr. Vikas Sapkal" with paper ID "IJPREMS50300028336" has published a paper entitled "AUTOMATED CONTROL AND PROTECTION IN INDUCTION MOTOR" in International Journal of Progressive Research in Engineering Management and Science (IJPREMS), Volume 05, Issue 03, March 2025



Managing Editor
IJPREMS Journal

www.ijprems.com



Indexing services:



.....



**International Journal of Progressive Research in
Engineering Management and Science**
(Int Peer Reviewed Journal)

Ref: I/Certificate/Volume 05/Issue 03 /50300028336

Impact Factor: 7.001

DOI: <https://www.doi.org/10.58257/IJPREMS39047>

e-ISSN: 2583-1062

Date: 18/03/2025

Certificate of Publication

This is to certify that author "Mr. Vijay Dokhale" with paper ID "IJPREMS50300028336" has published a paper entitled "AUTOMATED CONTROL AND PROTECTION IN INDUCTION MOTOR" in International Journal of Progressive Research in Engineering Management and Science (IJPREMS), Volume 05, Issue 03, March 2025



Managing Editor
IJPREMS Journal

www.ijprems.com



Indexing services:



.....



**International Journal of Progressive Research in
Engineering Management and Science**
(Int Peer Reviewed Journal)

Ref: I/Certificate/Volume 05/Issue 03 /50300028336

Impact Factor: 7.001

DOI: <https://www.doi.org/10.58257/IJPREMS39047>

e-ISSN: 2583-1062

Date: 18/03/2025

Certificate of Publication

This is to certify that author "Mr. Ganesh Kakde" with paper ID "IJPREMS50300028336" has published a paper entitled "AUTOMATED CONTROL AND PROTECTION IN INDUCTION MOTOR" in International Journal of Progressive Research in Engineering Management and Science (IJPREMS), Volume 05, Issue 03, March 2025



Managing Editor
IJPREMS Journal

www.ijprems.com



Indexing services:



.....



**International Journal of Progressive Research in
Engineering Management and Science**
(Int Peer Reviewed Journal)

Ref: I/Certificate/Volume 05/Issue 03 /50300028336

Impact Factor: 7.001

DOI: <https://www.doi.org/10.58257/IJPREMS39047>

e-ISSN: 2583-1062

Date: 18/03/2025

Certificate of Publication

This is to certify that author "Mr. Aniket Phirke" with paper ID "IJPREMS50300028336" has published a paper entitled "AUTOMATED CONTROL AND PROTECTION IN INDUCTION MOTOR" in International Journal of Progressive Research in Engineering Management and Science (IJPREMS), Volume 05, Issue 03, March 2025



Managing Editor
IJPREMS Journal

www.ijprems.com



Indexing services:



.....

CERTIFICATES



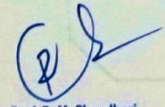
 **PROJECT Xpo 2K25**
NATIONAL LEVEL PROJECT COMPETITION

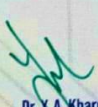
Certificate

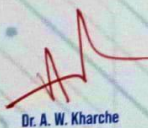
This Certificate is Proudly Presented to
Mr. / Miss. Arpit Sanjay Badhe
of padm. Dr. V. B. K. C. O. E. Malkapur
for participation and secured position: participated.
at National Level Technical Event "Project Competition" on 05th April 2025.

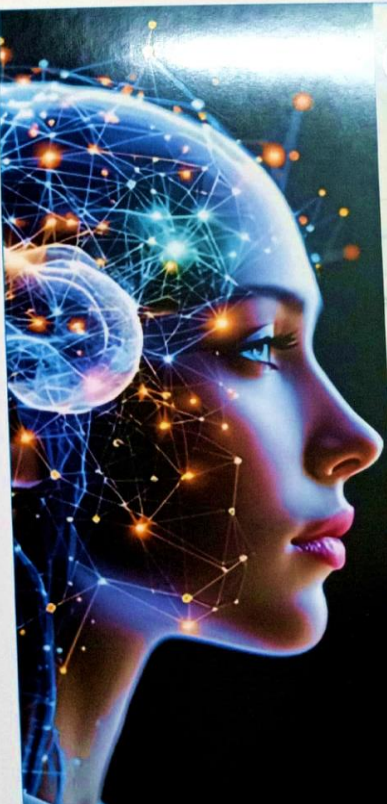
ORGANIZED BY
LSSBM'S
PADM. DR. V. B. KOLTE COLLEGE OF ENGINEERING, MALKAPUR
MUKTAINAGAR ROAD, MALKAPUR, DIST- BULDANA-443101
(NAAC & NBA Accredited, Approved by AICTE - New Delhi, Government of Maharashtra, DTE and Affiliated to S.G.B. Amravati University)

 Prof. D. P. Kharat
Coordinator

 Prof. R. M. Choudhari
Dean IQAC

 Dr. Y. A. Kharche
Dean (Administration)

 Dr. A. W. Kharche
Principal



 **PROJECT Xpo 2K25**
NATIONAL LEVEL PROJECT COMPETITION

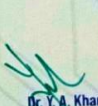
Certificate

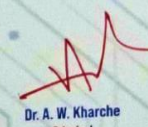
This Certificate is Proudly Presented to
Mr. / Miss. Aniket Vilay phiske
of padm. Dr. V. B. K. C. O. E. Malkapur
for participation and secured position: participant.
at National Level Technical Event "Project Competition" on 05th April 2025.

ORGANIZED BY
LSSBM'S
PADM. DR. V. B. KOLTE COLLEGE OF ENGINEERING, MALKAPUR
MUKTAINAGAR ROAD, MALKAPUR, DIST- BULDANA-443101
(NAAC & NBA Accredited, Approved by AICTE - New Delhi, Government of Maharashtra, DTE and Affiliated to S.G.B. Amravati University)

 Prof. D. P. Kharat
Coordinator

 Prof. R. M. Choudhari
Dean IQAC

 Dr. Y. A. Kharche
Dean (Administration)

 Dr. A. W. Kharche
Principal





REFERENCE

1. Bocker, J., & Mathapati, S. (2007, May). State of the art of induction motor control. In 2007 IEEE International Electric Machines & Drives Conference (Vol. 2, pp. 1459-1464). IEEE.
2. Jacobina, C. B., de Rossiter Correa, M. B., da Silva, E. C., & Lima, A. M. N. (1999). Induction motor drive system for low-power applications. *IEEE Transactions on Industry Applications*, 35(1), 52-61.
3. Glowacz, A., & Glowacz, Z. (2017). Diagnosis of the three-phase induction motor using thermal imaging. *Infrared physics & technology*, 81, 7-16.
4. Lewis, C. (2002, July). The advanced induction motor. In *IEEE Power Engineering Society Summer Meeting*, (Vol. 1, pp. 250-253). IEEE.
5. Mamdouh, M., Abido, M. A., & Hamouz, Z. (2018). Weighting Factor Selection Techniques for Predictive Torque Control of Induction Motor Drives: A Comparison Study. *Arabian Journal for Science and Engineering*, 43(2), 433– 445. <https://doi.org/10.1007/s13369-017-2842-2>
6. Journal, I. (n.d.). A REVIEW PAPER ON VARIABLE FREQUENCY DRIVE.
7. Bonet-Jara, J., Quijano-Lopez, A., Morinigo-Sotelo, D., & Pons-Llinares, J. (2021). Sensorless speed estimation for the diagnosis of induction motors via mcsa. Review and commercial devices analysis†. *Sensors*, 21(15), 1–35. <https://doi.org/10.3390/s21155037>
8. Jannati, M., Anbaran, S. A., Asgari, S. H., Goh, W. Y., Monadi, A., Aziz, M. J. A., & Idris, N. R. N. (2017). A review on Variable Speed Control techniques for efficient control of Single-Phase Induction Motors: Evolution, classification, comparison. *Renewable and Sustainable Energy Reviews*, 75(April), 1306–1319. <https://doi.org/10.1016/j.rser.2016.11.115>
9. Ouanjli, N. El, Derouich, A., Ghzizal, A. El, Motahhir, S., Chebabhi, A., Mourabit, Y. El, & Taoussi, M. (2019). Modern improvement techniques of direct torque control for induction motor drives-A review. *Protection and Control of Modern Power Systems*, 4(1). <https://doi.org/10.1186/s41601-019-0125-5>
10. Hannan, M. A., Ali, J. A., Mohamed, A., & Hussain, A. (2018). Optimization techniques to enhance the performance of induction motor drives: A review. *Renewable and Sustainable Energy Reviews*, 81(September 2016), 1611–1626. <https://doi.org/10.1016/j.rser.2017.05.240>

11. Subasri, R., Meenakumari, R., Panchal, H., Suresh, M., Priya, V., Ashokkumar, R., & Sadasivuni, K. K. (2020). Comparison of BPN, RBFN and wavelet neural network in induction motor modelling for speed estimation.
12. International Journal of Ambient Energy, 0(0), 1–6.
<https://doi.org/10.1080/01430750.2020.1817779>
13. MTalla, J., Leu, V. Q., Smidl, V., & Peroutka, Z. (2018). Adaptive speed control of induction motor drive with inaccurate model. IEEE Transactions on Industrial Electronics, 65(11), 8532–8542. <https://doi.org/10.1109/TIE.2018.2811362>
14. Sobhi, S.; Reshadi, M.; Zarft, N.; Terheide, A.; Dick, S. Condition Monitoring and Fault Detection in Small Induction Motors Using Machine Learning Algorithms. Information 2023, 14, 329. <https://doi.org/10.3390/info14060329>
15. Gudiño-Ochoa, A.; Jalomo-Cuevas, J.; Molinar-Solís, J.E.; Ochoa-Ornelas, R. Analysis of Interharmonics Generation in Induction Motors Driven by Variable Frequency Drives and AC Choppers. Energies 2023, 16, 5538. <https://doi.org/10.3390/en16145538>
16. Chang, Y., Yan, H., Huang, W., Quan, R., & Zhang, Y. (2023). A novel starting method with reactive power compensation for induction motors. IET Power Electronics, 16(3), 402-412.
17. Priya V. Kale, Amit M. Kale, Nikhil R. Kamdi, Prof. Ankita A. Yeotikar, “Protection of induction motor using classical method”, International Research Journal of Engineering and Technology, Volume: 04 Issue: 01 | Jan -2017.
18. Ganapathy Subramanian, “Fault Detection and Protection of Induction Motors Using Sensors”, IEEE IFET College Of Engineering Villupuram III-BE, Department.
19. M. E. H. Benbouzid, “Bibliography on induction motors faults detection and diagnosis”, IEEE Trans. Energy Convers., vol. 14, no. 4, pp. 1065-1074, Dec. 1999.
20. Shital S. Kalbande, Vaishali M. Kamble , Priyanka G. Kale , Prof. Ankita A. Yeotikar, “Protection and monitoring of three phase induction motor from over voltage, under voltage, single phasing, phase reversal, and overheating review”, IRJET, Volume: 04 Issue: 01 | Jan - 2017.
21. Shweta hugar, Basavaraj Amarapur, “Protection of Induction Motor using Microcontroller”, IJETAE, Volume 6, Issue 5, May 2016.
22. I. Colak, H. Celik, I. Sefa, and S. Demirbas, —On line protection system for induction motors, || Energy Convers. Manage vol. 46, no. 17, pp. 2773– 2786, 2005.

23. Y. Zhongming and W. Bin, —A review on induction motor online fault diagnosis,|| in 3rd Int. Power Electron. Motion Control Conf. (PIEMC 2000), vol. 3, pp. 1353–1358.
24. W. A. Farag and M. I. Kamel, —Microprocessor-based protection system for three-phase induction motors,| Electr. Mach. Power Syst., vol. 27, pp. 453– 464, 1999.
25. Chetan Borse, Akshay Pandhare and Randhir kumar, “Plc based induction motor starting and protection”,International Journal of Engineering Research and General Science Volume 3, Issue 2, March-April, 2015.
26. Sujith John Mathew, B.Hemalatha, “Fault Identification and Protection of Induction Motor using PLC and SCADA”, International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering, Vol. 3, Issue 4, April 2014.
27. D. Kornack and P. Rakic, “Cell Proliferation whereas not Neurogenesis in Adult Primate cerebral cortex,” Science, vol.294, Dec. 2001, pp. 2127-2130,doi:10.1126/science.1065467.
29. W.H. Kersting, "Causes and effects of single-phasing induction motors", IEEE Transactions on business Applications, vol. 41, no. 6, pp. 1499-1505, Dec. 2005.
30. P.E. Dame Joan Sutherland and T.A. Short, "Effect of Single-Phase Reclosing on Industrial Loads", business Applications Conference 2006. forty first Annual Meeting, vol. 5, pp. 2636-2644, 8–12 Oct. 2006.
31. G.C. Stone, M. Sasic, D. Dunn and I. Culbert, "Recent issues practised with motor and generator windings", 56th AnnualPetroleum and industry Conference PCIC 2009, pp. 1-9, 14–16 Sept. 2009 .
32. Rakesh Makode¹ , Vibha Nikose² , Zeeshan Ahmad³ , Adarsh Ramteke⁴ , Nidhi Paunikar⁵ , e⁴ , Nidhi Paunikar⁵ , Vishal Dashriya⁶ , Prof. D.D. Dhawale “IOT Based Automation and Monitoring of Induction Motor” National Conference on “Recent Trends in Electrical Engineering” NCRTEE-19 Organized by Pri- yadarshini College of Engineering, Nagpur-440019 In- ternational Journal of Innovations in Engineering and Science, Vol. 4, No.7, 2019, e-ISSN: 2456-346
33. Quijano-Lopez, A., Morinigo-Sotelo, D., & Pons-Llinares, J. (2021). Sensorless speed estimation for the diagnosis of induction motors via mcsa. Review and commercial devices analysis

34. B.Hemalatha, “Fault Identification and Protection of Induction Motor using PLC and SCADA”, International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering, Vol. 3, Issue 4, April 2014.
35. Akshay Pandhare and Randhir kumar, Chetan Borse, “Plc based induction motor starting and protection”, International Journal of Engineering Research and General Science Volume 3, Issue 2, March-April, 2015.
36. I. Sefa, and S. Demirbas, I. Colak, H. Celik, —On line protection system for induction motors, || Energy Convers. Manage vol. 46, no. 17, pp. 2773– 2786, 2005.
37. Vaishali M. Kamble , Priyanka G. Kale , Shital S. Kalbande, Prof. Ankita A. Yeotikar, “ Protection and monitoring of three phase induction motor from over voltage, under voltage, single phasing, phase reversal, and overheating review”, IRJET, Volume: 04 Issue: 01 | Jan - 2017.
38. MTalla, J., Leu, V. Q., Smidl, V., & Peroutka, Z. (2018). Adaptive speed control of induction motor drive with inaccurate model. IEEE Transactions on Industrial Electronics, 65(11), 8532–8542. <https://doi.org/10.1109/TIE.2018.2811362>
39. Glowacz, Z. (2017). Glowacz, A., Diagnosis of the three-phase induction motor using thermal imaging. Infrared physics & technology, 81, 7-16.

CODING

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>

// Initialize LCD display
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Bluetooth module
SoftwareSerial BTSerial(10, 11);

// Pin Definitions
const int relayMotor = 2;
const int resetPin = 8;
const int vibrationSensorPin = 6;
const int tempSensorPin = A3;
const int currentSensorPin = A1;
const int buzzerPin = 7;
const int capacitorBank = 5;
const int inductorBank = 4;
#define IR_SENSOR 3

// Fault States
bool faultDetected = false;
String faultMessage = "";

// Sensor Parameters
const float overCurrentLimit = 1.5;
const float overTempLimit = 70.0;
const float seriesResistor = 10000.0;
const float nominalResistance = 10000.0;
```

```

const float bCoefficient = 3950.0;
const float nominalTemperature = 25.0;
const float voltage = 240.0; // Fixed voltage for power factor calculation

volatile int pulseCount = 0;
unsigned long previousMillis = 0;
const int interval = 1000; // 1 second interval

void rpmCounter() {
    pulseCount++; // Increment count when IR sensor detects a change
}

void setup() {
    pinMode(relayMotor, OUTPUT);
    pinMode(resetPin, INPUT_PULLUP);
    pinMode(vibrationSensorPin, INPUT);
    pinMode(buzzerPin, OUTPUT);
    pinMode(capacitorBank, OUTPUT);
    pinMode(inductorBank, OUTPUT);
    pinMode(IR_SENSOR, INPUT);
    attachInterrupt(digitalPinToInterrupt(IR_SENSOR), rpmCounter, FALLING);
    digitalWrite(relayMotor, LOW);
    digitalWrite(buzzerPin, LOW);
    digitalWrite(capacitorBank, LOW);
    digitalWrite(inductorBank, LOW);
    lcd.init();
    lcd.backlight();
    BTSerial.begin(9600);
    Serial.begin(9600);
    delay(2000); // *Increased stabilization delay at startup*
}

void loop() {

```



```

float motorCurrent = readCurrent();
float motorTemp = readTemperature();
bool vibrationDetected = readVibration();
float powerFactor = calculatePowerFactor(motorCurrent);
int rpm = 0;
unsigned long currentMillis = millis();
// *RPM Calculation with Delay for Stability*
if (currentMillis - previousMillis >= interval) {
    previousMillis = currentMillis;
    rpm = pulseCount * 60; // Convert pulse count to RPM
    pulseCount = 0; // Reset pulse count
}
// *Update LCD Display*
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("I:"); lcd.print(motorCurrent, 2);
lcd.print("A T:"); lcd.print(motorTemp, 1);
lcd.print("C");
lcd.setCursor(0, 1);
lcd.print("Vib:"); lcd.print(vibrationDetected ? "Yes" : "No");
lcd.print(" RPM:"); lcd.print(rpm);
// *Send Data to Mobile via Bluetooth*
BTSerial.print("Current: "); BTSerial.println(motorCurrent);
BTSerial.print("Temp: "); BTSerial.println(motorTemp);
BTSerial.print("Vibration: "); BTSerial.println(vibrationDetected ? "Yes" : "No");
BTSerial.print("Power Factor: "); BTSerial.println(powerFactor);
BTSerial.print("RPM: "); BTSerial.println(rpm);
adjustPowerFactor(powerFactor);
// *Fault Detection with Stabilization*
if (motorCurrent > overCurrentLimit) {
    delay(500); // Stabilization
    if (motorCurrent > overCurrentLimit) triggerFault("Overload Fault");
}

```

```

    }
    if (motorTemp > overTempLimit) {
        delay(500); // Stabilization
        if (motorTemp > overTempLimit) triggerFault("Over Temp Fault");
    }
    if (vibrationDetected) {
        delay(500); // Stabilization
        if (readVibration()) triggerFault("Vibration Detected");
    }
    // *Bluetooth Command Handling*
    if (BTSerial.available()) {
        char command = BTSerial.read();
        if (command == '1') digitalWrite(relayMotor, HIGH);
        if (command == '0') digitalWrite(relayMotor, LOW);
    }
    // *Handle Fault Condition*
    if (faultDetected) {
        digitalWrite(buzzerPin, HIGH);
        lcd.clear();
        lcd.print("Fault: Motor OFF");
        lcd.setCursor(0, 1);
        lcd.print(faultMessage);

        if (digitalRead(resetPin) == LOW) resetSystem();
    }
}

float readCurrent() {
    delay(300); // *Added delay for stable current readings*
    int sensorValue = analogRead(currentSensorPin);
    float voltage = sensorValue * (5.0 / 1023.0);
    float current = (voltage - 2.5) / 0.185; // Assuming ACS712 sensor
    return (current < 0.02) ? 0 : current;
}

```

```

}
float readTemperature() {
    delay(300); // *Delay for stabilization*
    int adcValue = analogRead(tempSensorPin);
    // *Avoid invalid readings (start from a normal temp)*
    if (adcValue == 0 || adcValue == 1023) return 25.0;
    float resistance = seriesResistor * ((float)adcValue / (1023 - adcValue));
    float steinhart = log(resistance / nominalResistance) / bCoefficient;
    steinhart += 1.0 / (nominalTemperature + 273.15);
    float temperature = (1.0 / steinhart) - 273.15;

    // *Invert the temperature reading correctly*
    temperature = 50 - (temperature - 25);

    // *Ensure gradual temperature adjustment*
    static float lastTemp = 25.0;
    temperature = (lastTemp * 0.7) + (temperature * 0.3);
    lastTemp = temperature;
    return temperature;
}
bool readVibration() {
    delay(300); // *Added delay for stable vibration readings*
    return digitalRead(vibrationSensorPin) == HIGH;
}

```